Logical Thinking

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Logical Thinking

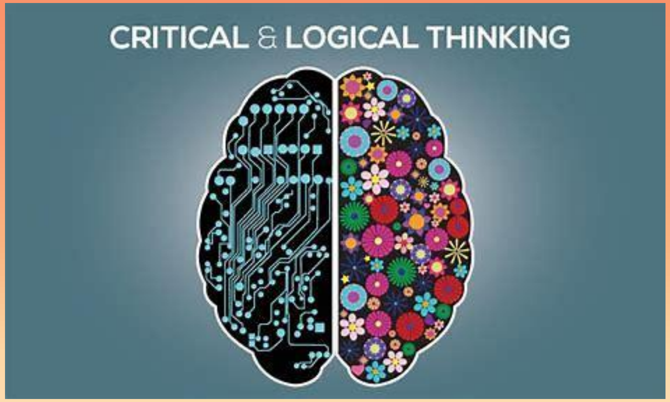
Question: How can you divide 10 oranges between 11 people evenly?

Logical Thinking

The definition of logical thinking is analyzing a situation or problem using reason and coming up with potential solutions.

Difference between Critical and logical thinking

- Critical thinking is closely related to logical thinking. It involves the questioning of data, beliefs, and information to make a reasoned conclusion or decision.



Logical Thinking

Difference between Critical and logical thinking

- On the one hand, logical thinking is pretty straightforward.
- It's a method of thinking that uses logic or analysis of information to evaluate a situation.

Logical Thinking vs Critical Thinking

Logical thinking is often straightforward. It's based on facts and figures. Critical thinking, on the other hand, involves questioning those facts and analyzing whether they're accurate or relevant. Both approaches are essential for making well-rounded decisions, but they serve different purposes.

Example: Let's say you're buying a new phone. Logical thinking would compare the prices, features, and reviews. Critical thinking, on the other hand, might ask, 'Do I really need a new phone right now? Is the newest model worth the money, or could I wait for a better deal?'

Logical Thinking vs Critical Thinking

1. Process vs. Reflection:

Logical thinking follows established rules and processes to reach a solution.
Example: You need to decide which of two job offers to accept. You create a pros and cons list, giving each option a score.

Critical thinking, on the other hand, reflects on the information and considers whether those rules and processes are valid or appropriate.
Example: A critical thinker might look at the same job offers and question the factors being considered. “Am I focusing too much

Logical Thinking vs Critical Thinking

2. Objective vs. Analytical:

Logical thinking tends to be more objective and fact-driven.
Example: In math, you use logical steps to solve an equation. You’re working within a well-defined set of rules, and there is usually one correct answer.

Critical thinking is more analytical and evaluative, questioning the context and assumptions behind those facts.
Example: When reading a news article, a critical thinker might question the bias of the source, the accuracy of the data, and the motives behind the story.

Logical Thinking vs Critical Thinking

3. Solving vs. Understanding:

Logical thinking is problem-solving-oriented. It focuses on solving the task at hand using predefined methods.
Example: In a game of chess, you logically calculate your next move by considering your options and choosing the one that will yield the best outcome based on known rules.

Critical thinking involves understanding the broader implications and considering multiple perspectives.
Example: In the same chess game, a critical thinker might evaluate not only the next move but also the strategy of the opponent. Why did they make a certain move? Are they trying to lure you into a trap? What

Logical Thinking vs Critical Thinking

4. Application of Rules vs. Questioning Rules:

Logical thinking sticks to applying existing rules to reach a solution.
Example: In a traffic situation, you follow traffic rules to drive safely (e.g., stopping at a red light, yielding to pedestrians).

Critical thinking might question the rules themselves.
Example: In a traffic scenario, a critical thinker might question why the rule exists in a certain situation. "Why is there a stop sign here when there's no intersection? Is this rule outdated or ineffective for current traffic conditions?"

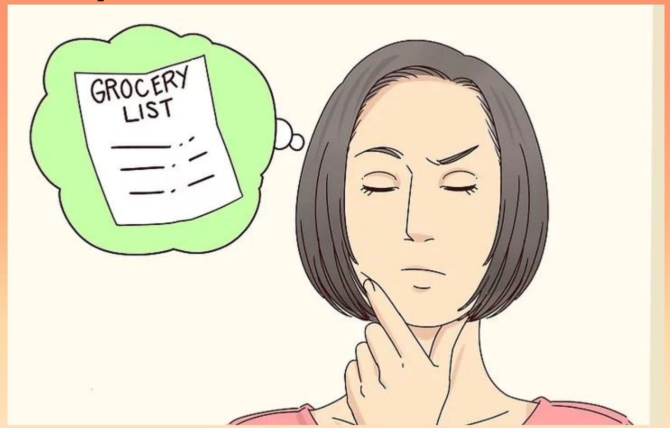
Logical Thinking: Your company has a policy to handle customer complaints by escalating them to a supervisor after three failed attempts to resolve the issue. You follow this rule.
Critical Thinking: You might ask whether this policy is actually effective. Are customers satisfied with this process, or would a different approach work better?

Logical Thinking: A person may follow a structured diet plan, eating certain foods in certain portions to lose weight based on their calorie content.
Critical Thinking: You question whether the diet is sustainable in the long term. Is this plan aligned with your lifestyle? Are there better, healthier options?

How to Think Logically

Exercising Your Test, your recall. Your brain, like any other body part, improves with exercise. A great way to give your brain a workout is to test your recall. Throughout the day, see how many details of a given moment, list, or task you can remember.

- Try to memorize small things each day. Write a grocery list and commit it to memory. Memorize a small passage from a poem or book. Wait an hour and see how much you can recall of what you've committed to memory.

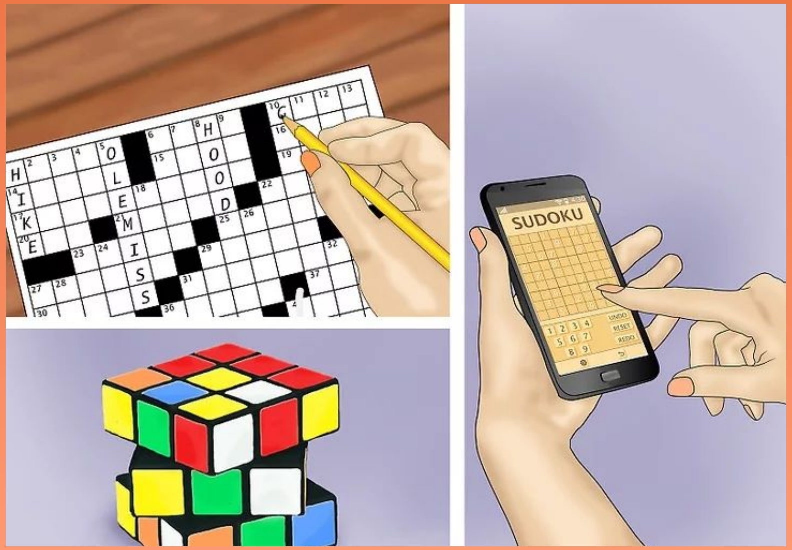


How to Think Logically

Do crossword puzzles. The benefit of crossword puzzles on the mind is well documented. Crossword puzzles force you to push your brain slightly beyond its capabilities, which causes the regrowth of brain neurons.

How to Think Logically

Do crossword puzzles. The benefit of This increases your overall brain power and can promote more sound, logical thinking. Pick up a crossword puzzle book from a local bookstore or do your local newspaper's crossword each morning.

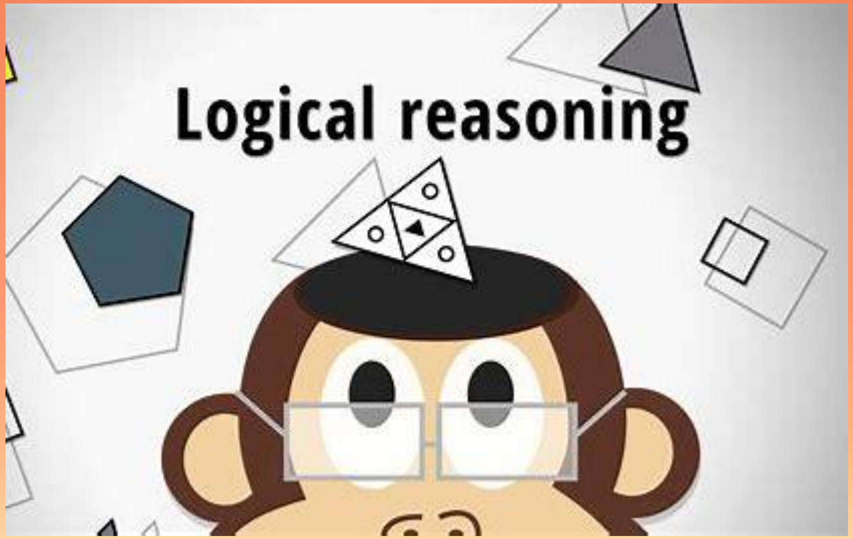


Learn a new talent. Learning new abilities requires a lot of logical thinking. From devising strategies that help you learn to undertake challenging tasks, you use logic and strategy to acquire new skills.

Try taking up some of the following activities to boost your logical thinking skills:

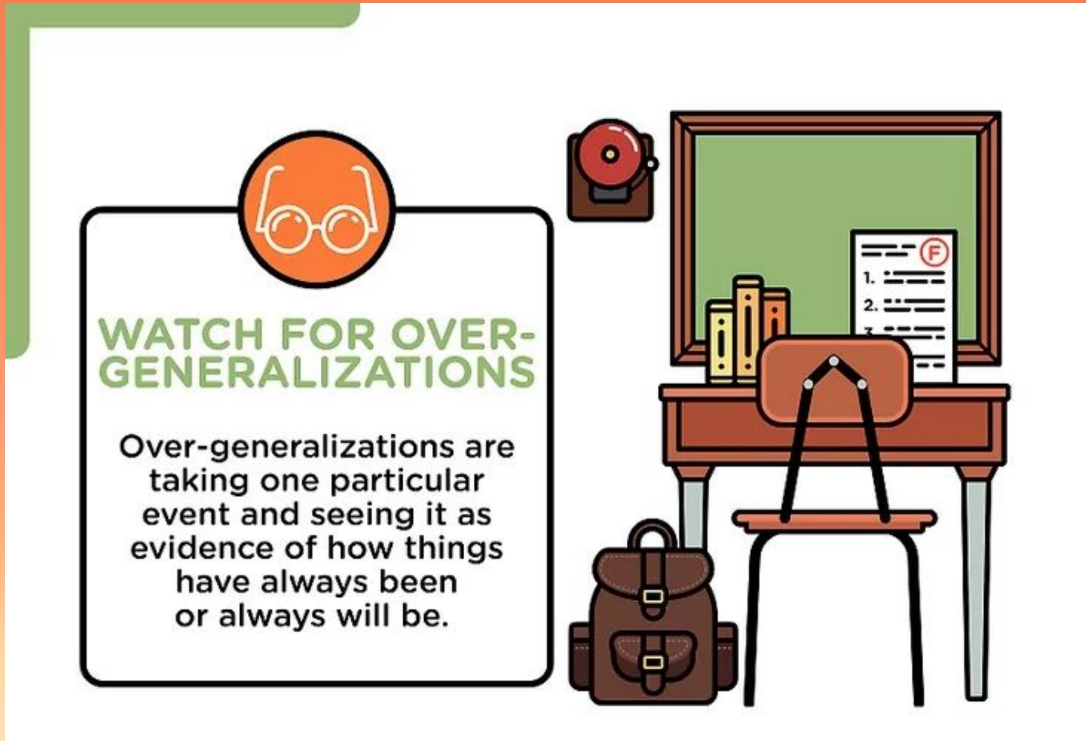
- Learn to play instruments.
- Learn to draw or paint.
- Learn to speak a foreign language.
- Learn to cook.

Pay attention to the purpose behind your actions. Each time you make a decision throughout the day, there is some purpose behind it. Given the hectic demands of day-to-day life, people sometimes lose track of the purpose and goals behind their actions.

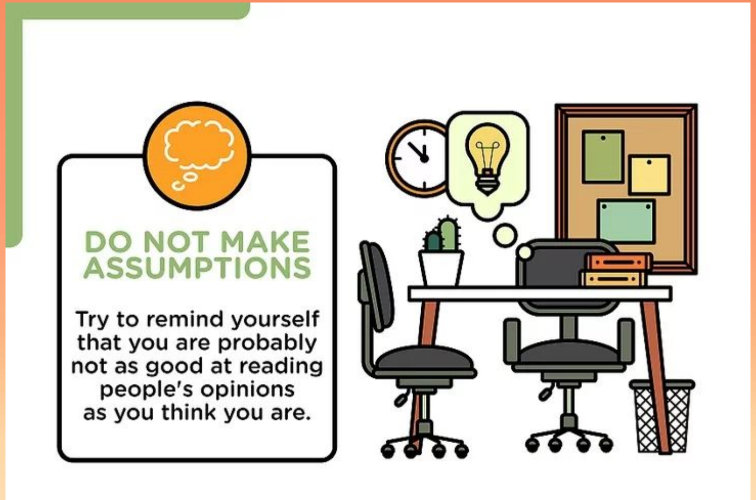


Focus on larger goals at work or school. Where do you want to be in five years? Two years? One year? How are you current actions serving this goal? Do your actions make sense reasonably given your larger purpose? Answering these questions can help you improve your reasoning skills.

Watch for over-generalizations. Many people over generalize without realizing it. This is toxic to rational thinking. Try to be aware of any over-generalizations you may make in day-to-day life.



Over-generalizations are taking one particular event and seeing it as evidence of how things have always been or always will be. For example, if you do bad on one test you may think, "I'm stupid and always fail at school." In making this statement, you're glossing other academic success you have had in the past in the light of one even.



Situation	Irrational Thought	Rational Thought
After a couple of dates that didn't lead to a second one, Jane thinks, "I'll always be alone."	*No one will ever be interested in dating me. There must be something wrong with me.*	*Not every date will lead to a second one, and that's okay. It doesn't define my entire dating experience or future relationships.*
Carla tried a new recipe, and it didn't turn out well.	*I'm a disaster in the kitchen and can't cook anything right.*	*One dish not turning out doesn't mean I'm a bad cook. I can learn and try again or make adjustments next time.*

Take two situations where you may have used **Overgeneralization.** Write down the irrational thoughts you had. Next write out what rational thoughts would be.

Situation	Irrational Thought	Rational Thought

Types of Logical Thinking:

Deductive reasoning begins with a broad truth (the major premise), such as the statement that all men are mortal. This is followed by the minor premise, a more specific statement, such as that Socrates is a man. A conclusion follows: Socrates is mortal.

Types of Logical Thinking:

In inductive reasoning, broad conclusions are drawn from specific observations; data leads to conclusions. If the data shows a tangible pattern, it will support a hypothesis. For example, having seen ten white swans, we could use inductive reasoning to conclude that all swans are white.

Observing that the sun rises every morning an always will.

Inductive

Using specific observations to form a general conclusion

VS

Deductive

Using a general premise to form a specific conclusion

YOUR DICTIONARY

Deductive Reasoning
Definition:
You start with a **general rule** or statement and apply it to a **specific case** to reach a certain conclusion.
Hint to remember:
👉 *General → Specific*
Example:
All humans need water.
Ali is a human.
→ Ali needs water.
Inductive Reasoning
Definition:
You start with **specific examples or observations** and make a **general conclusion** based on them.
Hint to remember:
👉 *Specific → General*
Ali, Sara, and Ahmed are humans, and all of them need water to survive.
→ Therefore, *probably* all humans need water. (Specific → General)

Deductive:
All cats have whiskers.
Milo is a cat.
→ Therefore, Milo has whiskers. (General → Specific)
Inductive:
Every cat I’ve seen has whiskers.
→ Probably, all cats have whiskers. (Specific → General)

Inductive Reasoning

- Moves from specific observations to a general conclusion.
- The conclusion is probable, not certain.
- Often used in science to make generalizations.

VS

Deductive Reasoning

- Moves from a general rule to a specific conclusion.
- The conclusion is certain if the premises are true.
- Often used in mathematics and logic to prove theorems.

Thank you!